Course code: ITA0605
course name: Machine Learning

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CO_2 - AT III - Analytical Q.

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1) A Genetic Algorithm is used to Maximize the function $f(x) = 2x+3$ for an initial population of values {1,3,5,7} Compute the fitness of each individual and select the top two candidates. perform a single point crossover to generate offspring and explain how much mutation can help avoid premature convergence

Given
$f(x) = 2x+3$

Individual	Fitness $f(x)$
1	$2(1)+3=5$
3	$2(3)+3=9$
5	$2(5)+3=13$
7	$2(7)+3=17$

the two highest fitness values are
$x=7 \rightarrow$ Fitness $=17$
$x=5 \rightarrow$ fitness $=13$

These are selected as parent chromosomes represent the parent in 3 bit binary
7=111
5=101

choose a crossover point after the first bit

parent 1: 1|11
parent 2: 1|01

swap the right parts
offspring 1: 1 01 = 101 = 5
offspring 2: 1 11 = 111 = 7

Offspring produced : 5 & 7

Fitness Values : 5, 9, 13, 17 , top two candidates : 7 & 5

offspring after single point crossover : 5 & 7

Mutation : Randomly alters genes to Maintain diversity and prevent premature convergence.

2) In a neural network, a neuron is trained using the delta learning rule. The initial weight is 0.6 the learning rate is 0.15 and input is 2.5. The actual output is 0.7, while the target output is 1 Calculate the error, determine the weight update and compute the new updated weight after one iteration.

Given,

Initial weight = $w = 0.6$ | Input $x = 2.5$

Learning rate , $\eta = 0.15$ | Actual output, $y = 0.7$

Target output, $t = 1$

Error = $t - y = 1 - 0.7 = 0.3$

$\boxed{\text{Error} = 0.3}$

Delta learning rule

$\Delta w = \eta \times (t - y) \times x$

$\Delta w = 0.15 \times 0.3 \times 2.5$

$\boxed{\Delta w = 0.1125}$

Weight update = 0.1125

New weight = ?

$w_{new} = w_{old} + \Delta w = 0.6 + 0.1125$

$\boxed{w_{new} = 0.7125}$

3) A simple perceptron model is used to classify whether a student passes or fails based on two input features, namely study hours and attendance. The weights associated with these inputs are 0.6 & 0.4 respectively, and the bias value is -0.5. For a given student with study hours equal to 2 & attendance equal to 1, compute the net input to the perceptron and apply the step activation function to determine the final output. Based on the results, conclude whether the student passes or fails.

Given

Study hours $(x_1) = 2$

Attendance $(x_2) = 1$

Weight $(w_1) = 0.6$

Weight $(w_2) = 0.4$

bias $(b) = -0.5$

Net input :-

$(0.6 \times 2) + (0.4 \times 1) - 0.5$

$= 1.2 + 0.4 - 0.5 = 1.1$

If net input $\geq 0 \rightarrow$ output $= 1$

otherwise $\rightarrow$ output $= 0$

Since $1.1 \geq 0$, output $= 1$

Conclusion : The student passes.

4) In a neural network, a single neuron is trained using the back propagation learning rule. The initial weight is 0.5 & the learning rate is 0.1 For an input of 2, the actual ouput produced by the neuron is 0.6, Whereas the target output is 1 Calculate the error in the output and determine the weight update using the delta rule Hence, compute the new updated weight after one iteration of training

Given
Initial weight = 0.5
learning rate (η) = 0.1
Input (x) = 2

Actual output = 0.6
Target output = 1

Error = 1 − 0.6 = $\boxed{0.4 \cdot \text{Error}}$

Δw = η × Error × Input
= 0.1 × 0.4 × 2 = $\boxed{0.08 = w_{update}}$

New weight
= 0.5 + 0.08 = $\boxed{0.58 = W_{new}}$

5) A deep learning model is developed for disease prediction and its performance is evaluated using a confusion matrix. The model yields 180 true positive, 150 true Negatives, 20 FP, 50 FN. using these values, compute accuracy, precision, sensitivity & specificity of the model

Based on the Calculated results, using discuss whether the model is reliable for medical diagnosis

Given $TP = 180$, $TN = 150$, $FP = 20$, $FN = 50$

Accuracy

$$= \frac{TP + TN}{TP + TN + FP + FN} = \frac{330}{400} = 82.5\%$$

precision

$$= \frac{180}{180 + 20} = \cancel{78.26}\ 90\%$$

Recall

$$= \frac{180}{180 + 50} = 78.26\%$$

$$\text{Specificity} = \frac{150}{150 + 20} = 88.24\%$$

The Model has good accuracy, precision & specificity, but recall is comparatively lower.